

Circumstellar Nebulae around Short-Period Eclipsing Binaries in the Small Magellanic Cloud

Yuka Terada¹, You-Hua Chu^{1,2}, Chien-Jia Lee³

1. National Sun Yat-sen University, Kaosiung, Taiwan, 2. Institute of Astronomy and Astrophysics, Academia Sinica, Taipei, Taiwan, 3. National Taiwan University, Taipei, Taiwan

INTRODUCTION

The Dark Energy Camera Magellanic Cloud Emission Line Survey (DeMCELS; Points et al. 2024) has revealed a rich population of ionized gas structures, from small nebulae associated with massive stars or evolved stars to supernova remnants, superbubbles, and supergiant shells. With its superb angular resolution, DeMCELS also revealed a large number of small nebulae (diameter < 5 pc) that are apparently associated with stars. We tentatively call them circumstellar nebulae (CSNe). While a fraction of these CSNe may be planetary nebulae, a significant number of them appear to be associated with eclipsing binaries (EBs). In our initial examination of the SMC, we have already identified 60 CSNe, among which 21 are associated with EBs.

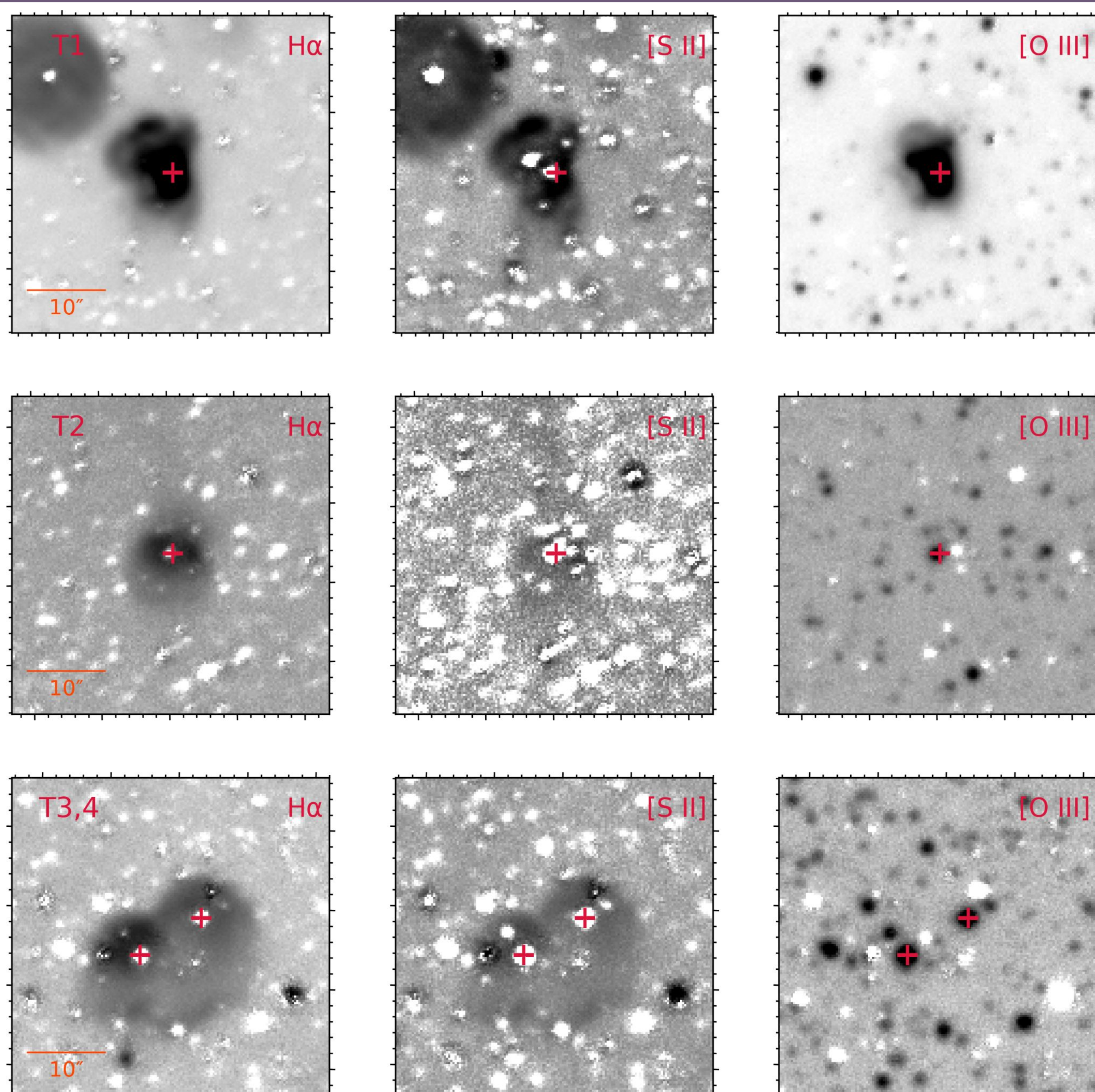
The origin of these nebulae is unclear. In particular, it remains unknown whether binary interaction can produce such structures, and what physical mechanisms drive mass ejection in these systems.

In this work, we explore the observational properties of EB-associated CSNe using multi-band imaging (H α , [S II], [O III]) and time-domain photometry from OGLE. This study represents an exploratory step toward understanding a potentially new class of nebulae associated with close binary evolution.

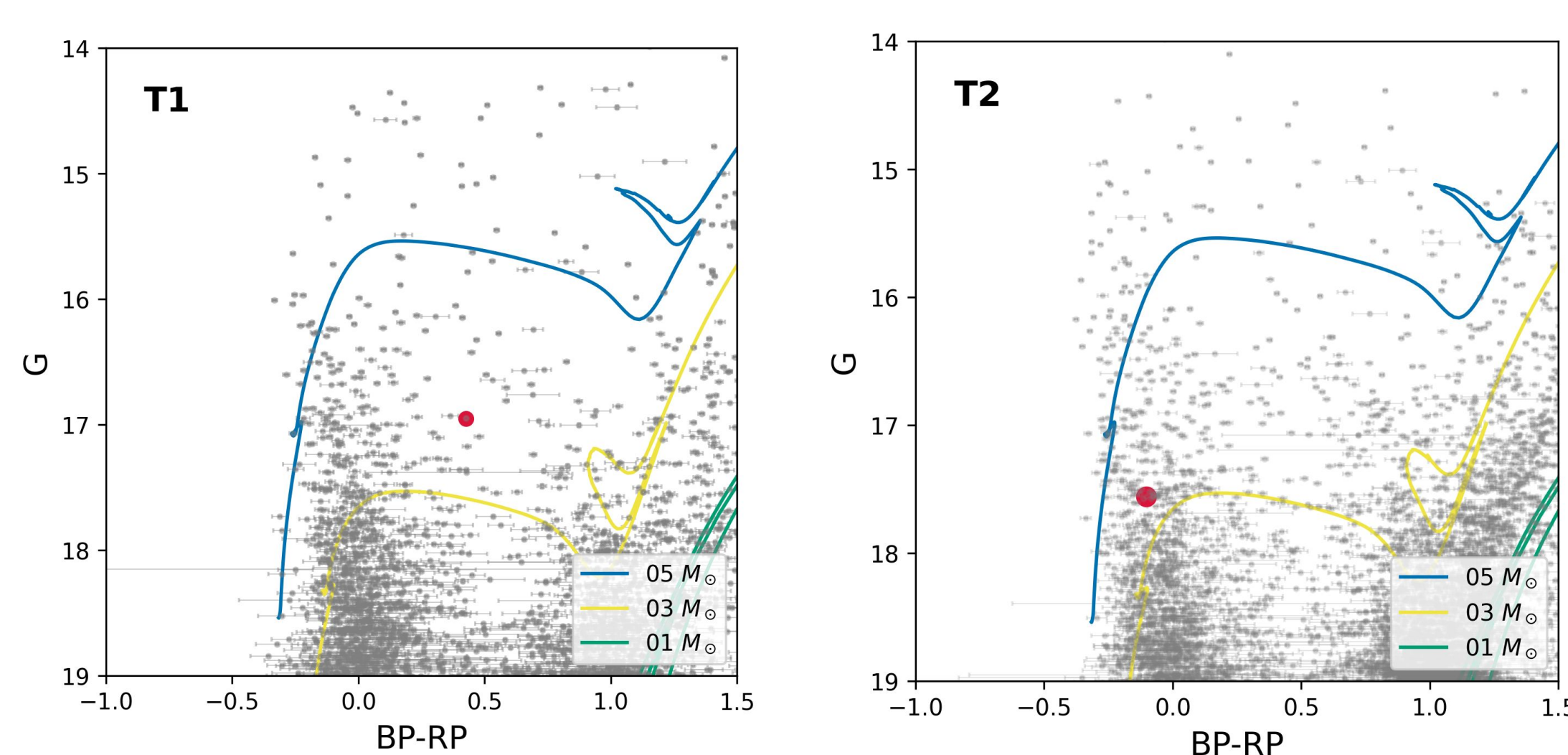
DATA & SAMPLES

- DeMCELS imaging (H α , [S II], [O III])
- Visual selection of CSN candidates
- Cross-match with OGLE and Gaia
- Identification of EBs from light curves
- Analysis of morphology and distributions

IMAGING RESULTS



- We present three representative EB nebulae to illustrate their diversity.
- Most EB nebulae (~80%) show H α emission with weak [S II] and no detectable [O III] (e.g., T2), and are typically compact without shell structure.
- Their detailed nebular morphologies may differ in shape, symmetry, and clumpiness.
- A small fraction shows [O III] emission (e.g., T1), indicating higher excitation conditions.
- An intriguing region contains two EB nebulae (T3,4) separated by ~9'' (~2.6 pc), comparable to the nebular size, suggesting a possible physical association.
- Overall nebular sizes range from ~5'' to ~25'', corresponding to spatial scales of a few parsecs.
- Gaia CMD (G_{BP}-G_{RP} vs G) constructed and overlaid with MIST evolutionary tracks (Dotter 2016; Choi et al. 2016; Dotter et al. 2026; Bauer et al. 2025).

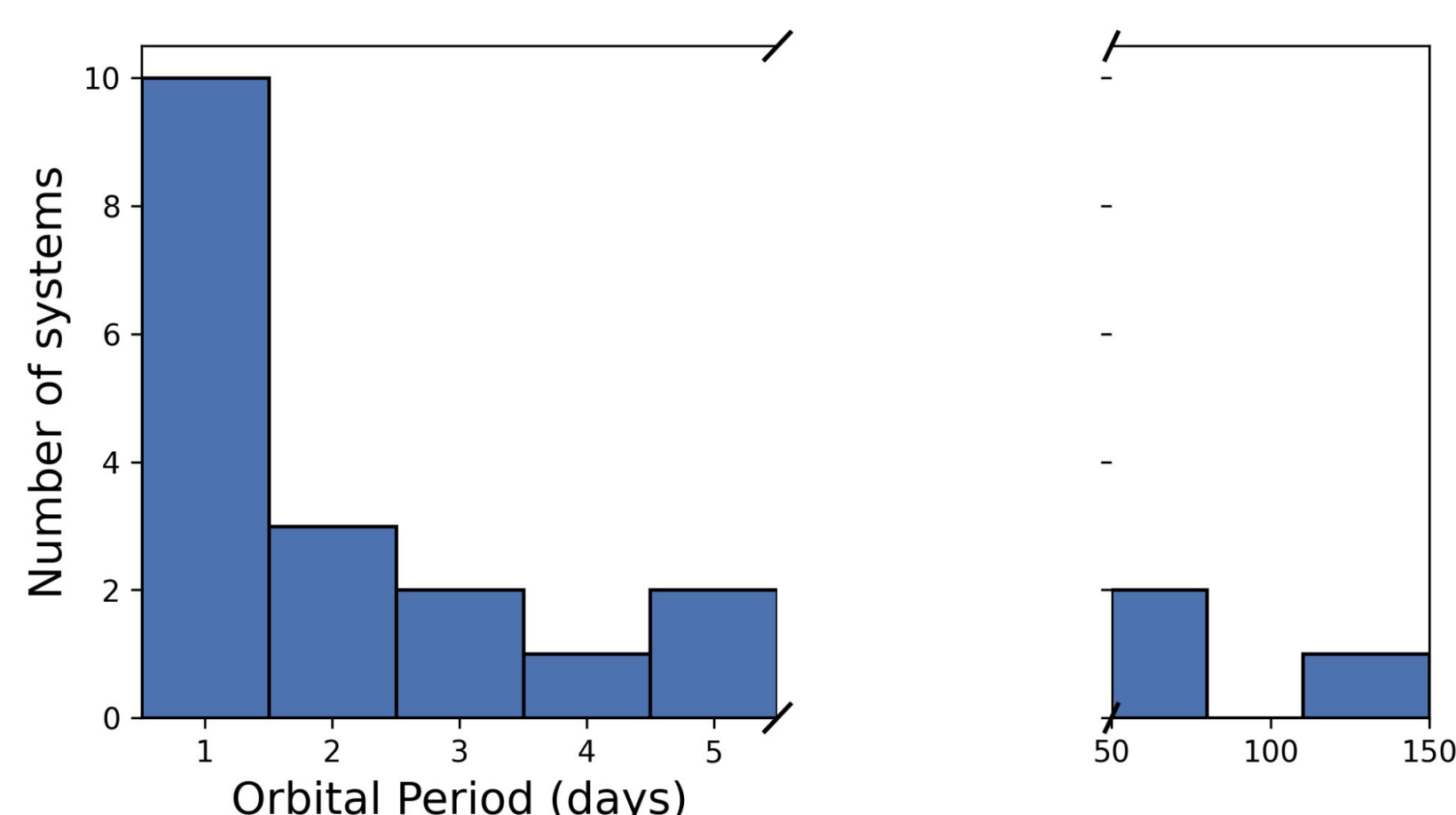


- No clear mass difference is seen between T1 and T2.
- The sample spans ~5–10 M $_{\odot}$, indicating a range of stellar masses.

PROPERTIES

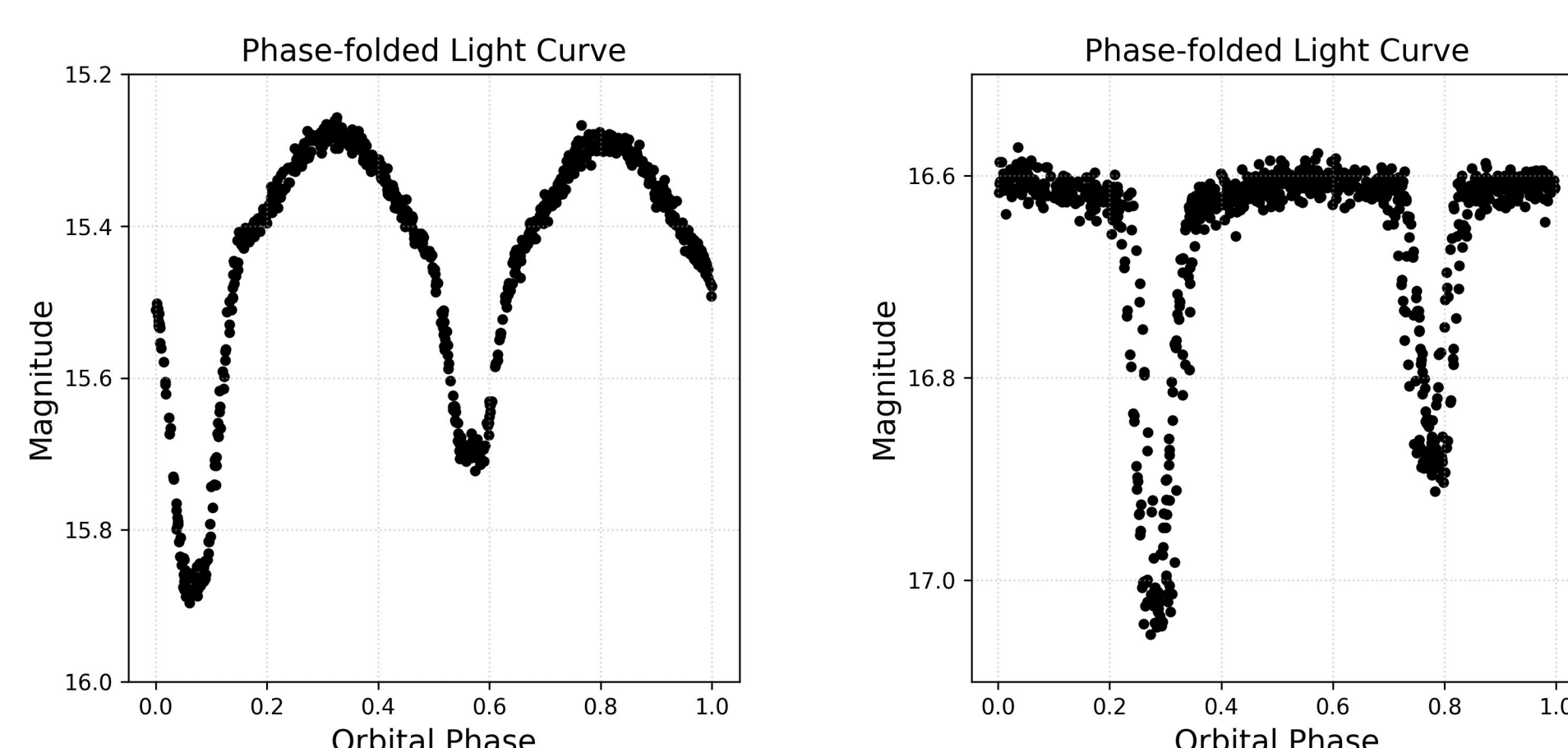
Period Distribution

- Orbital periods are compiled from OGLE (13) and Gaia (11); OGLE values are adopted when both are available.
- Most systems have short periods:
 - ~50% (8/18) have $P < 1.5$ days
 - 15/18 have $P < 5.5$ days
- This concentration at short periods suggests close binaries, likely undergoing or approaching interaction.



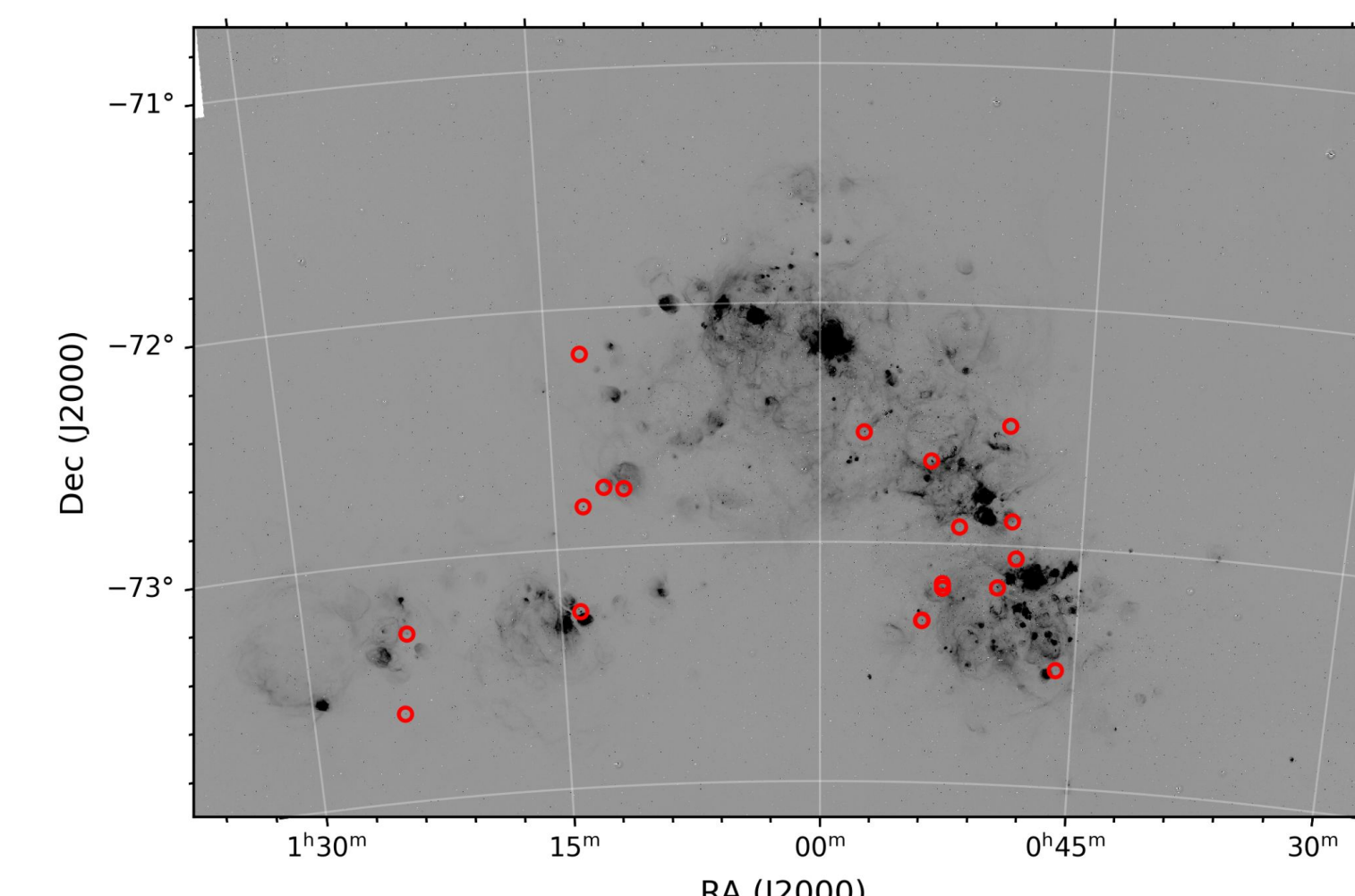
Light Curves

- Phase-folded OGLE light curves show diverse morphologies among EB-associated CSNe.
- Some systems exhibit strong ellipsoidal modulation with distorted baselines (left), indicating tidal interaction.
- Others show relatively flatter baselines despite similarly short periods (right).
- This diversity suggests a range of binary configurations and interaction stages.



Spatial Distribution

- EB-associated CSNe are marked on the SMC H α map; the survey is still ongoing and the current sample is incomplete.
- The detected systems tend to be located in regions of strong H α emission, suggesting a possible association with star-forming environments.

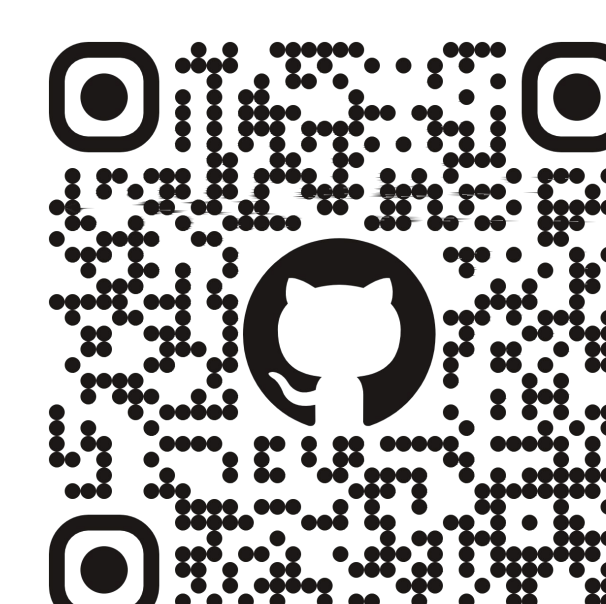


SUMMARY

- We identify a population of circumstellar nebulae (CSNe) associated with eclipsing binaries (EBs) in the SMC (60 CSNe, 21EB nebulae), suggesting a potentially new class of objects.
- Most systems have short orbital periods, indicating close binaries where tidal interaction and mass loss are likely important.
- Imaging and light curves reveal significant diversity in nebular morphology and variability, implying a range of binary configurations and evolutionary stages.
- The nebulae have typical sizes of a few parsecs and are often located in H α -bright regions, hinting at a connection with star-forming environments.
- These properties suggest multiple formation pathways, including binary interaction and common-envelope evolution.
- Such systems may provide links between close binary evolution, massive star formation, and supernova remnant progenitors.
- Ongoing work, including detailed light-curve modeling and collaboration on common-envelope and dust processes, will further clarify their origin.

REFERENCES

Points, S. D. 2024, ApJ, 974, 70
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